

MAINCRAFTS TECHNOLOGY

Cybersecurity & Ethical Hacking Internship

TASK 2: BUILD YOUR PERSONAL CYBERSECURITY LAB

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Task	Task 2 — Build Your Personal Cybersecurity Lab
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1. Objective

The objective of this task is to build a safe and isolated cybersecurity practice environment on a personal computer. This hands-on lab serves as the foundation for all future tasks such as vulnerability assessment, web application testing, exploitation practice, and incident response.

The following were created as part of this lab:

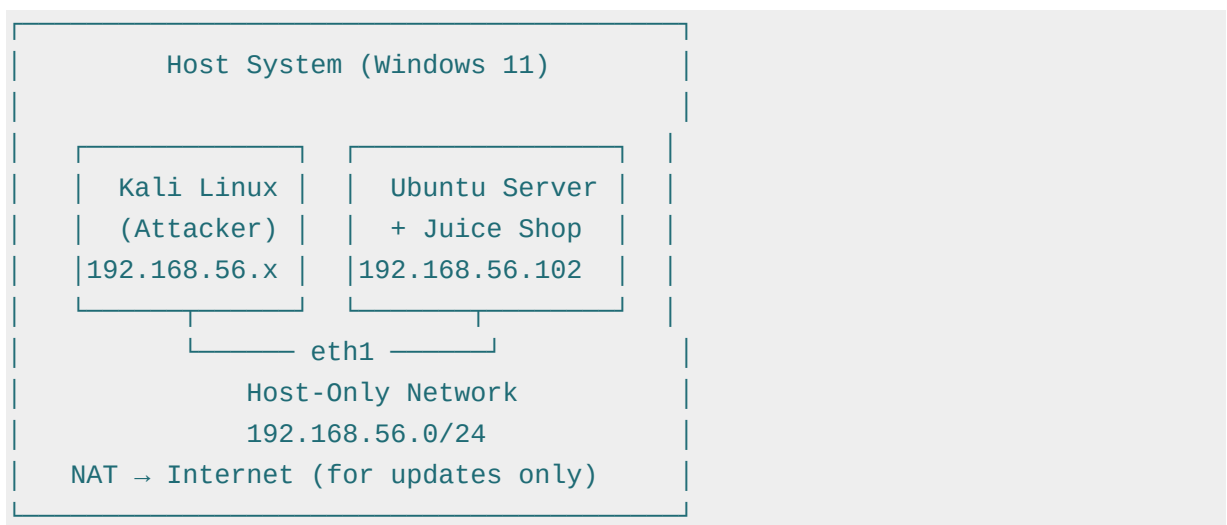
- 1 Attacker Machine — Kali Linux 2026.1 (VirtualBox VM)
- 1 Vulnerable Target Application — OWASP Juice Shop (Docker on Ubuntu Server 24.04)
- Proper Network Segmentation — NAT (internet) + Host-Only (lab traffic)

2. Host System Specifications

Device	ASUS Vivobook Go E1504FA
Processor	AMD Ryzen 5 7520U with Radeon Graphics @ 2.80 GHz
Installed RAM	16.0 GB (15.3 GB usable)
Storage	477 GB (178 GB free)
OS	Windows 11 (64-bit)
Virtualization	Oracle VirtualBox 7.x

3. Lab Architecture

The lab uses a two-VM setup with isolated Host-Only networking for attack traffic and NAT for internet access during setup.



4. Step-by-Step Implementation

Step 1: VirtualBox Installation & Network Setup

VirtualBox was installed on the Windows 11 host. A Host-Only network (vboxnet0) was created using VBoxManage CLI with subnet 192.168.56.0/24 to isolate lab traffic from the real network.

Step 2: Kali Linux VM Deployment

The pre-built Kali Linux 2026.1 VirtualBox image was downloaded, extracted using 7-Zip, and imported via File → Import Appliance. The VM was configured with 2 CPUs, 4 GB RAM, Adapter 1 as NAT, and Adapter 2 as Host-Only (vboxnet0).

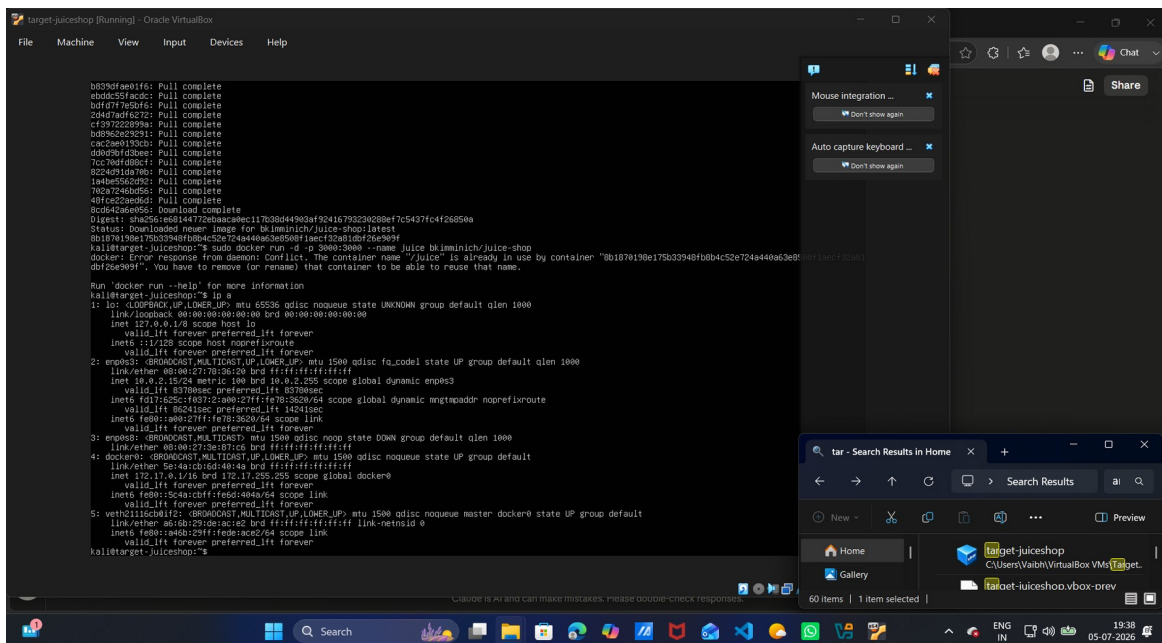
Step 3: Ubuntu Server + Juice Shop Deployment

Ubuntu Server 24.04.4 LTS was installed on a second VM with 1 CPU, 2 GB RAM, and 15 GB disk. After installation, Docker was installed and OWASP Juice Shop was deployed using the following commands:

```
sudo apt update && sudo apt install -y docker.io
sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
sudo ip addr add 192.168.56.102/24 dev enp0s8
```

Target VM IP Configuration

The target VM was assigned IP 192.168.56.102 on the Host-Only interface (enp0s8) as shown below:



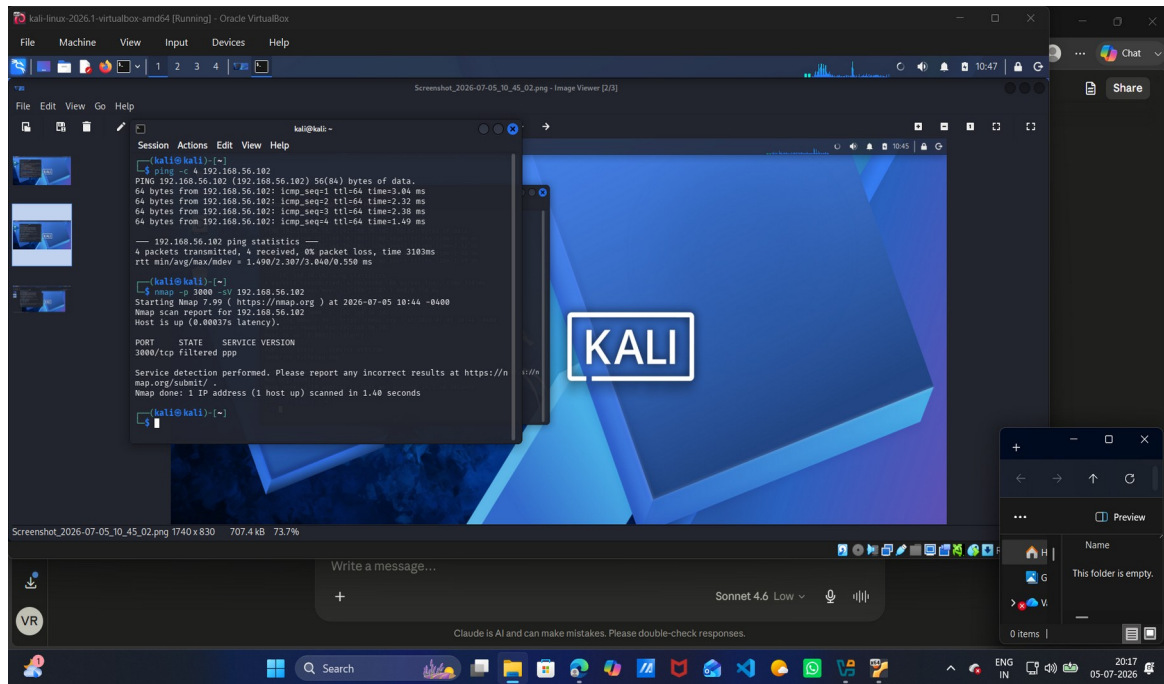
Screenshot 1: Target VM 'ip a' output showing 192.168.56.102 on enp0s8

5. Connectivity Verification

Ping & Nmap Scan from Kali

From the Kali attacker machine, connectivity to the target was verified using ping and Nmap port scan:

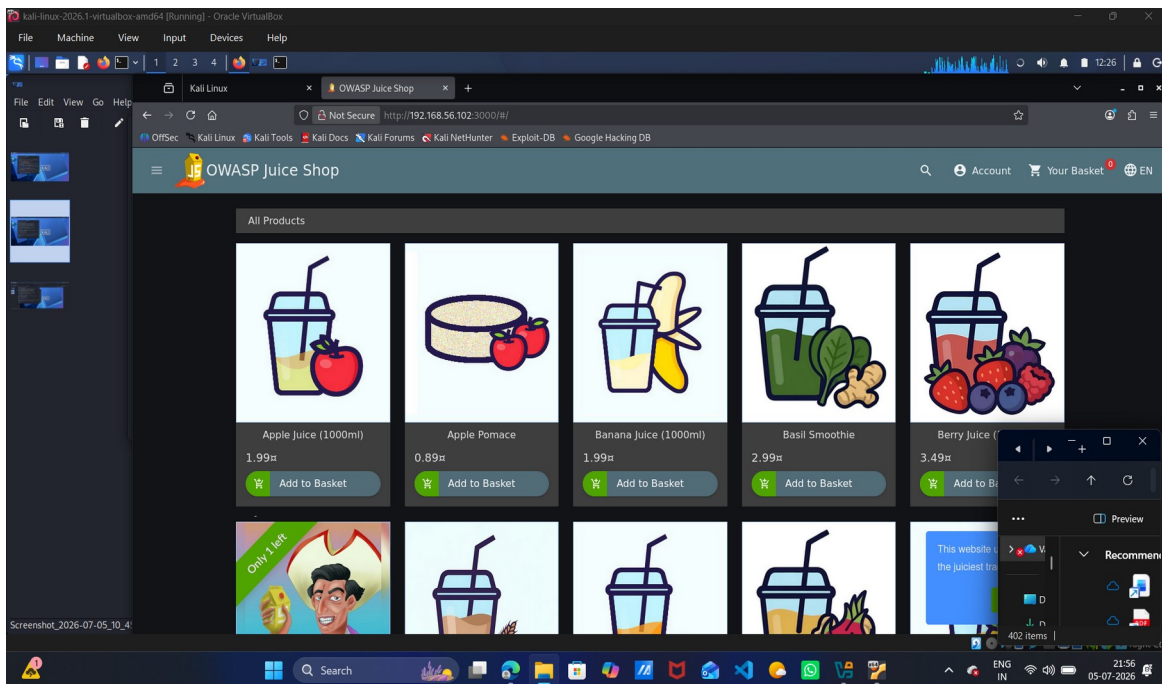
```
ping -c 4 192.168.56.102
nmap -p 3000 -sV 192.168.56.102
```



Screenshot 2: Ping showing 0% packet loss & Nmap confirming port 3000 open on target

6. OWASP Juice Shop Access

OWASP Juice Shop was successfully accessed from Kali's Firefox browser at <http://192.168.56.102:3000>, confirming the vulnerable web application is running and reachable from the attacker machine.

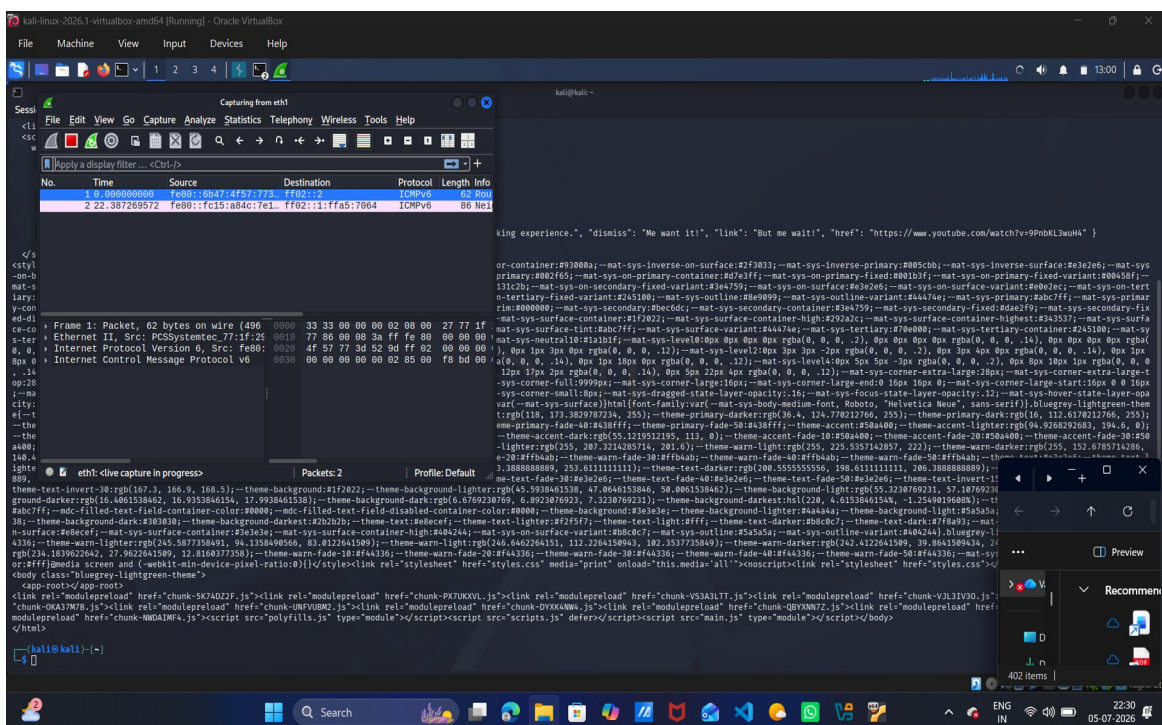


Screenshot 3: OWASP Juice Shop homepage loaded in Kali Firefox at <http://192.168.56.102:3000>

7. Tool Validation

Burp Suite — HTTP Traffic Interception

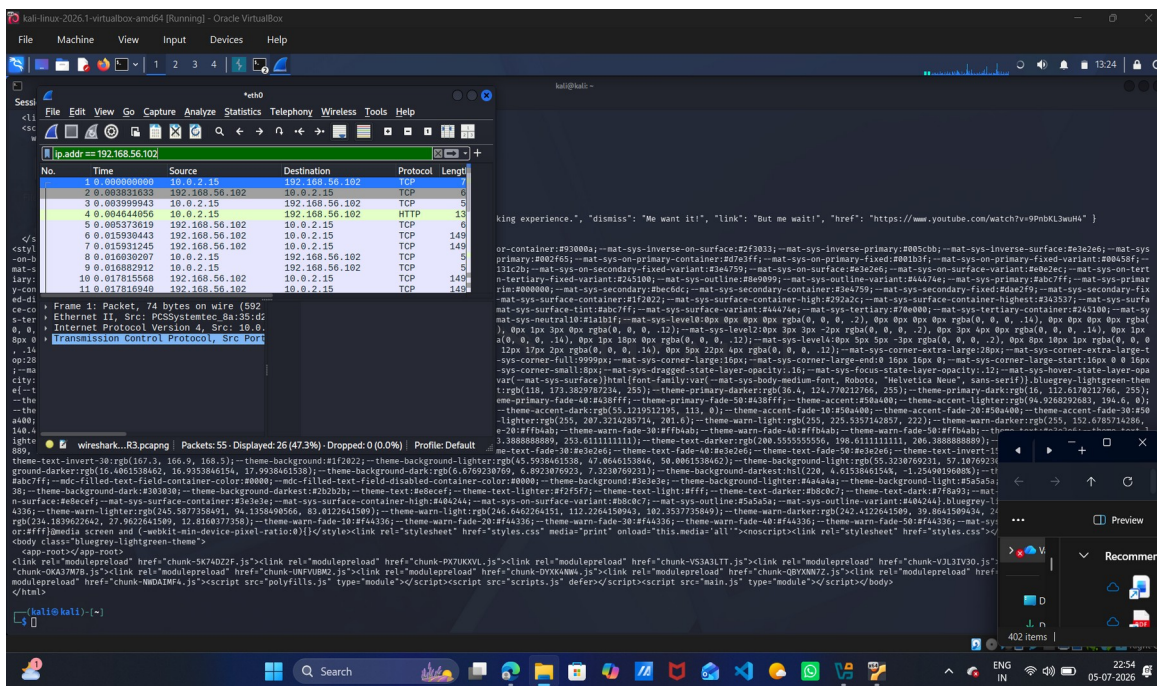
Burp Suite Community Edition was launched in Kali. Firefox was configured to proxy through 127.0.0.1:8080. HTTP requests to and from Juice Shop were successfully intercepted, demonstrating man-in-the-middle proxy capability.



Screenshot 4: Burp Suite HTTP history showing intercepted requests to 192.168.56.102:3000

Wireshark — Packet Capture & Analysis

Wireshark was used to capture live network traffic on the eth0 interface. A curl request was sent to the Juice Shop target. Packets were filtered using `ip.addr == 192.168.56.102`, confirming bidirectional TCP/HTTP traffic between Kali and the target.



Screenshot 5: Wireshark filtered packet capture showing TCP/HTTP traffic to 192.168.56.102

8. Learning Outcomes

Virtualization	Successfully ran two isolated VMs simultaneously for pentesting
Lab Isolation	NAT + Host-Only networking prevented real-world exposure
Web Target Setup	Deployed OWASP Juice Shop via Docker on Ubuntu Server
Basic Networking	Configured NAT vs Host-Only, subnets, manual IP assignment
Tool Awareness	Validated Nmap, Burp Suite, and Wireshark in lab environment

9. Conclusion

The personal cybersecurity lab was successfully built and fully validated. The two-VM architecture with Kali Linux as the attacker and Ubuntu Server running OWASP Juice Shop as the target, connected via Host-Only networking, provides a safe and isolated environment for all future pentesting tasks. All required tools —

Nmap, Burp Suite, Wireshark, Docker, and VirtualBox — were confirmed working. This lab will serve as the foundation for upcoming tasks including vulnerability assessment, web application exploitation, and incident response.

— *End of Report* —